

Section I

The Science and Physiology of Stretching



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1 My Personal Motivation for Stretching

The year is 1972. A 15-year-old, 5 foot, 10-inch (1.78 meters), 175-lb (79.4 kg) fullback takes his stance three yards behind the quarterback. This grade 11 fullback playing junior high school football is bigger and stronger than most of the offensive linemen blocking for him as well as most of the young adolescent opponents who will try to tackle him. With the ball snapped on the second “hut”, the quarterback rotates and hands the ball off to the fullback who is accelerating to an expected opening between the tight end and tackle. As the tight end cross blocks upon the defensive end and the offensive tackle pulls out against the linebacker, a sliver of daylight appears. I lower my shoulder and plunge through that hole. Arms reach out from the partially blocked defensive lineman and linebacker but they are not strong enough to slow me down. A strong side defensive back at about 140 pounds (63.5 kg) moves in for the tackle. However, with the momentum of my greater mass and the velocity attained after an 8-yard sprint, the defensive back is trampled and I cut sharply to the sidelines. After covering about 20 yards, the defensive safety catches me from behind and trips me up.

The next year, I am the starting fullback for the senior high school team. I have grown $\frac{3}{4}$ of an inch (2 cm) and now weigh 185 lbs (84 kg). We win the regional high school championship. Dave Behm (The Truck) and Dan (Crazy Legs) Murphy make a great one-two punch. My predominant empire is inside between the two ends. Dives and off-tackle plays are my bread and butter. If I get a decent block and get into the defensive backfield, I can use my size advantage, my balance, and my signature move: hit the opponent at full throttle to either knock him down or spin immediately after contact so that the enemy cannot easily grab me and pull me down. Murphy’s territory is outside the ends with sweeps and pitches, as he is lighter and much swifter than I and can often outspurt the defence.

In the last year of high school, there is no increase in height but I continue to fill out, expanding to 198 lbs (90 kg). A number of Canadian universities attempt to recruit me and I decide to stay home to play with the University of Ottawa Gee-Gees, who had lost that year in the national semi-final game. However, as the next year approaches, I realize that my chances of getting into a game with that team are slim to none. The starting fullback is Neil

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Lumsden, a 235 lb (106 kg) behemoth with decent speed, great balance, strength, and power. He will set Canadian university rushing and touchdown records that will remain untouched for a couple of decades and he will subsequently establish a long career in the professional Canadian Football League (CFL). The back-up is Mike Murphy, another talented fullback at just under 230 lbs (104 kg) who in the following year will lead the nation in rushing yards and also have a firm career in the CFL. I decide to play for the city's junior football team (Ottawa Sooners) and wait for Lumsden to move onto the CFL. Lumsden and Murphy were archetype fullbacks, massive, strong, and powerful. Relatively, I filled that description in high school but my growth pattern started to plateau, such that I was around 205 lb (93 kg) when I joined the university football team in my second year of university. I spent my second year of university primarily blocking for the burly Mike Murphy and moved into the fullback position in my third academic year. I was lucky enough to inherit a strong offensive line and with my slashing, spinning style I could often break a few tackles and make a major gain (Figure 1.1). With my lack of breakaway running speed, I was typically caught from behind by a fleet footed defensive back. For a professional running back, my size was more typical of tailbacks or halfbacks depending on your terminology. These backs typified by legends like Walter Payton had very good to great speed that would allow them to burst into the open field



Figure 1.1 David Behm (32) caught “again” from behind

and outsprint the opposition. Unfortunately, I was built like a tailback but with the speed of a fullback. I needed to get faster if I wanted to continue my career after university.

Sprinting speed is a simple combination of stride rate (frequency) and stride length (7). Stride rate is very difficult to modify as it is generally related to your genetic profile of fast-twitch to slow-twitch muscle fibre composition. Pick the right parents, who hopefully will pass on a higher percentage of fast-twitch fibres and you will be able to move your legs back and forth (stride rate) much quicker than someone with a greater percentage of slow-twitch fibres. Hence, you will have a high stride rate. This is the most important factor in sprinting speed. Unfortunately, it seems that I did not pick the right parents! Thus, I was left with trying to enhance the second sprinting factor: stride length. With stronger more powerful legs, it should be possible to explode off the ground and cover greater distances with each stride. With this in mind, I worked faithfully on my strength, such that I could squat over 500 lbs (225 kg) and bench press around 350 lb (160 kg). As a university student in physical education in the late 1970s, I was taught that by increasing my flexibility I would improve performance (increase stride length, decrease resistance to stride movements) and decrease the chances for injury.

With these pearls of wisdom in mind, I also worked diligently on my stretching so that I could eventually perform a front split. Conventional wisdom of the time indicated that with higher levels of flexibility there would be less resistance to movement and an increased efficiency of movement. Thus, with my improved power and flexibility my stride length should have been tremendous. Like Superman, I should have been leaping over tall buildings in a single bound. However, there must have been some kryptonite in my diet since, while my sprinting speed did improve marginally, nobody ever mistook me for the legendary Walter Payton (or Superman!).

Well if I did not get much faster, then my flexibility should have decreased my chances for injury. In my second year of university football, I took the ball on a draw play (quarterback fakes a pass and then gives the ball to the running back) and burst through a gap in the line. Quickly a linebacker exploded from the left accelerating his helmet into my shoulder. I tried to absorb the hit and I bounced off the hit and continued for another 12 yards till I was, as usual, caught from behind by a defensive back. Upon getting up I noticed my clavicle was apparently redirected towards my back and no longer attached to my scapula. When I returned to the sidelines I was informed that I had subluxated my acromio-clavicular joint. I was out for the season.

With aggressive rehabilitation and off-season training I was ready for my third year of university football. In a game against McGill University on artificial turf, I caught a flare pass and sprinted wide. As I planted my right foot to move up field I was hit low on the left side by a linebacker and simultaneously hit high from the right by a defensive back. My planted foot could not move or slide on the artificial turf and the ligaments were torn resulting in a third-degree ankle sprain. It did not seem that my extensive

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stretching programme had provided me the protection I sought from musculotendinous or ligamentous injuries nor did it provide me with significantly better athletic performance (improved speed). Did my physical education professors of the 1970s really know what they were talking about? The paradigm of stretching and flexibility has experienced a number of shifts in the past few decades.

It is now about 50 years later (2024) and I am no longer a young athlete but I still try to maintain my fitness (Figure 1.2). Although I am officially a senior citizen (>65 years), I have not lost my competitive nature and still seek to improve my athletic skills in tennis and ice hockey (my arthritic knees make squash more difficult now!) so I can compete with and beat those youngsters (anybody under 50 years old) (Figure 1.3). However, my preparations and warm-ups have been altered by our research findings on flexibility and pre-activity preparations. As we now know that resistance training can provide similar increases in range of motion as static stretching (1), there is no longer a need to stretch before lifting weights. However, I still include static stretching (less than 60 seconds per muscle group: 3, 4, 5, 6) before I play tennis or ice hockey. Periodically if I really feel stiff, I might add foam rolling (2) to my regimen.

Hopefully the second edition of this book will update and help establish the facts about flexibility as they are presently known and burst the myths. It is the



Figure 1.2 Old David Behm trying to maintain fitness and fighting the ravages of ageing (65 years old). Possible positive effects of functional work, resistance training, competitive sport, and maybe stretching?



Figure 1.3 Old David Behm still competing in as many sports as possible. Did stretching contribute to my relative athletic success as a senior citizen?

objective of this book to provide you with the most up to date research on flexibility enhancing exercises and devices as well as explaining the physiological mechanisms underlying different types of flexibility-improving exercises/devices and then provide you with suggestions for appropriate activity programmes to increase your range of motion.

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